

THE STRUCTURE OF MATHEMATICAL REALITY

Phase III — Chapter 7

Groups: The Language of Symmetry

Assignment

Self-Study Curriculum

Choose a system you understand well (a biological system, a data pipeline, a physical process). Identify its symmetries, the transformations under which some property of the system is unchanged. Be specific about what property and what transformations. Then answer:

1. Do these symmetries form a group? (Check each axiom.)
2. Are there subgroups? (Partial symmetries?)
3. What does the quotient by a subgroup correspond to concretely in the system?

Present this as a worked example with explicit verification. If the symmetries do *not* form a group, explain which axiom fails and what that failure means structurally about the system.